

■ NASA 3D Models Verification Checklist (v3 - Complete)

Project: *Echoes in the Dust* (2026 NASA Space Apps Challenge - Challenge 1)
Repository Targets: Official NASA 3D Resources Catalog (science.nasa.gov/3d-resources/) & NASA JPL GitHub Repositories

1. Core Campaign Spacecraft & Landers

Hardware Name	Location / Mission	NASA 3D Status	Official Asset URL / Repository
Apollo Lunar Module (<i>Eagle</i>)	Moon (Sea of Tranquillity)	■ FOUND	NASA 3D Resource
Opportunity & Spirit Rovers	Mars (Meridiani Planum / Gusev)	■ FOUND	NASA 3D Resource
Pioneer 10 Interstellar Probe	Deep Space (Taurus trajectory)	■ FOUND	NASA 3D Resource
Viking Lander 1 & 2	Mars (Chryse & Utopia Planitia)	■ FOUND	NASA 3D Resource
Ingenuity Mars Helicopter	Mars (Jezero Crater)	■ FOUND	NASA 3D Resource
InSight Lander	Mars (Elysium Planitia)	■ FOUND	NASA 3D Resource
Kepler Space Telescope	Deep Space (Earth-trailing)	■ FOUND	NASA 3D Resource
Spitzer Space Telescope	Deep Space (Earth-trailing)	■ FOUND	NASA 3D Resource
Dawn Spacecraft	Asteroid Belt (Ceres)	■ FOUND	NASA 3D Resource
LADEE Moon Orbiter	Moon (Far side impact)	■ FOUND	NASA 3D Resource

2. Secondary & Partial Hardware (EDL, Mobility & Surface Systems)

Hardware Category	Location / Mission	NASA CAD/Model Status	Official Asset URL / Repository
Sky Crane Stage & EDL Debris	Mars (Jezero / Gale Crater)	■ VERIFIED	NASA JPL Mars 2020 URDF Models
Parachute & Backshell Assemblies	Mars (Atmosphere / Surface)	■ VERIFIED	NASA JPL Mars 2020 URDF Models
Rocker-Bogie Chassis & Suspension	Mars (Sojourner / MER / M2020)	■ VERIFIED	NASA JPL Open Source Rover CAD
Lunar Roving Vehicle (LRV)	Moon (Apollo 15, 16, 17)	■ VERIFIED	NASA Central 3D Resources
Apollo ALSEP & Retroreflectors	Moon (Apollo 11–17 Sites)	■ VERIFIED	Apollo 15 ALSEP & LRV Resources
Central 3D Repository Archive	Multi-Mission Asset Archive	■ VERIFIED	NASA Central 3D Repository

3. Historical Telemetry & PDS Reference Hardware

Hardware Name	Location / Mission	Primary Data Type	Source / Archive Notes
Surveyor Probes (1–7)	Moon (Ocean of Storms, etc.)	PDS Surface Imaging	ALSJ & PDS Geosciences Archives
LCROSS Centaur Stage	Moon (Cabeus Crater)	Spectrometry Data	PDS Geosciences Node Data
GRAIL Probes (Ebb & Flow)	Moon (Sally Ride Crater)	Gravity Mapping Data	Planetary Data System Archives
Pioneer Venus Probes	Venus Atmosphere	Atmospheric Entry Logs	NSSDCA Master Catalog
NEAR Shoemaker	Asteroid 433 Eros	Asteroid 3D Shape Model	NASA 3D Resources Hub

- Verification Summary Stats:
- Total Hardware Categories Cataloged: 21
 - Official 3D / CAD / URDF Models Verified: 16 (76.2%)
 - PDS & Scientific Telemetry References: 5 (23.8%)